NP-hard reduction

Problem. A town is laid out as a grid, and a car can travel between points (x_1, y_1) and (x_2, y_2) in $|x_1 - x_2| + |y_1 - y_2|$ units of time; denote this distance by d. You are given m identical cars, all waiting at a depot o at time 0, and a list of n trip requests $r_1, \ldots, r_n$, where each request $r_i = (s_i, w_i, t_i)$ asks for a passenger to be picked up at s_i and dropped off at w_i, and is made at time t_i. All requests are known in advance.

Each car can carry only one passenger at a time and serves its trips one after another. If a car finishes its previous trip at location w' at time e (initially $w' = o$ and $e = 0$), it can pick up the passenger for request r_i at time

$$p_i = \max\{t_i,\ e + d(w', s_i)\},$$

and drops them off at time $p_i + d(s_i, w_i)$. The *waiting time* of request r_i is $p_i - t_i$.

The objective is to assign every request to a car, together with the order in which each car serves its requests, so as to minimise the maximum waiting time

$$W_{\max} = \max_{1 \le i \le n} (p_i - t_i).$$

Theorem. Minimising the maximum waiting time is NP-hard.

Proof. We reduce the makespan problem to our problem. Specifically, we use the makespan problem on m identical machines, restricted to instances whose processing times $q_1, \ldots, q_n$ satisfy $q_{\max} \le 2q_{\min}$. This restricted problem is still NP-hard. Given an instance of 3-Partition, which is NP-complete (Garey & Johnson, 1979), consisting of positive integers $a_1, \ldots, a_{3m}$ with $\sum_j a_j = mB$ and $B/4 < a_j < B/2$ for all j, set $q_j = a_j$. Since the total processing time is mB, the jobs can be scheduled on m machines with makespan at most B if and only if they can be split into m sets each summing to exactly B, and any such set contains exactly three numbers since $B/4 < a_j < B/2$. Moreover, $q_{\max} < B/2 < 2q_{\min}$.

Consider the following instance of our problem. We have n trips and m cars. The depot is at $o = (0, 0)$, where all cars start at time 0. Trip j starts at $s_j = (q_j, 0)$, and all end locations are at the car depot, so

$$r_j = (s_j,\ o,\ t_j), \qquad \text{where } t_j = q_{\max} - q_j \ge 0.$$

This instance can be constructed in polynomial time.
Since every trip ends at the depot, each car is at the depot whenever it begins a new trip. If a car begins trip j at time S_j, it reaches s_j at time $S_j + q_j$. Since $q_{\max} \le 2q_j$, we have

$$S_j + q_j \ \ge\ q_j \ \ge\ q_{\max} - q_j \ =\ t_j,$$

so the trip has always been requested by the time the car arrives. Hence the car picks up the passenger at time $p_j = S_j + q_j$ without waiting, and returns to the depot at time $C_j = S_j + 2q_j$.

Then this is just the makespan problem, where each car is a machine and each trip is a job, and the processing time of trip j is the time to go to the start

location of the trip and then go back to the depot, $2q_j$. The waiting time of trip j is

$$p_j - t_j \;=\; (C_j - q_j) - (q_{\max} - q_j) \;=\; C_j - q_{\max}.$$

Therefore, for any schedule, the maximum waiting time is $\max_j C_j - q_{\max}$, which is the makespan of the corresponding machine schedule minus the constant $q_{\max}$. So a schedule minimises the maximum waiting time if and only if it minimises the makespan. Since the processing times are $2q_j$, the minimum maximum waiting time is $W^*_{\max} = 2C^* - q_{\max}$, where C^* is the minimum makespan of the original instance.

Hence, given an algorithm for our problem, we can compute $C^* = (W^*_{\max} + q_{\max})/2$, and the transformation takes polynomial time. Therefore, since our problem can solve the makespan problem, it must be NP-hard. □